Appendix 15 for Math 632: 2025 Winter Notes

Chunye Yang

April 17, 2026

In this part of the note series we return to the notion of **flatness**, which is a concept that is very algebraic at first glance. We could have discussed this way before, but a number of examples that shows that this algebraic notion is in fact geometric are very connected to the notion of cohomology. For example, if $\pi : X \to Y$ is a projective flat morphism over a connected Noetherian base Y, then we will see that a lot of numerial invaraints of fibers are constant, including dimension and numbers interpretable in terms of Euler characteristics:

1. The Hilbert polynomial.
2. The degree in projective space.
3. As a special case of 2, if we have a finite morphism π, then it's degree is constant. Recovering our previous fact that the degree of a finite morphism between regular curves are constatn.
4. The arithmetic genus.
5. The degree of a line bundle if the fiber is a curve.
6. Intersections of divisors and line bundles.

Therefore, this flatness seems to be the right notion we need to describe a nice varying family over a base. This is indeed case, and you will see in the next chapter that flatness will be the starting point of the definition of a **moduli space**. For example, you will see, as an example, the **Hilbert scheme** of $\mathbb{P}^n$ parametrizes closed subschemes of $\mathbb{P}^n$. Maps from a scheme B to the Hilbert scheme correspond to finitely presented closed subschemes of $\mathbb{P}^n_B$ flat over B. Another example is the moduli space of smooth projective curves, which is defined by universal property that morphisms into this moduli space correspond to projective flat finitely presented families whose geometric fibers are smooth curves. (However, this modulis space doesn't exist.)

Flatness is also very central in **deformation theory**, which is the key to understand schemes or other geometric objects can deform. Finally, if you further read the notes on etale cohomology, flatness is central in the **faithfully flat descent theory** which tells you how to glue in Grothendieck topologies in the same way as we do in the usual Zariski topolygy.

Finally, since flatness implies a lot of numerical invariants coming from cohomology is constant, you should not be surprised that flatness can also be characterized cohomologically. We will do this first before we head to the pleasant geometric implications coming out of flatness. Besides from the general criterion we give to check flatness, you should also pay attention to how flatness is checked over special kinds of rings, such as over an integral domain, a PID, a DVR and a local ring.

1 Easier facts

Some facts about flatness are easy and immediate, but there are facts out there that requires more work. We begin with the low-hanging fruits of flatness first.

Recall that if M is an A-module, we say M is **flat over** A if $M \otimes_A -$ is an exact functor from Mod_A to itself. We say a ring map $A \to B$ is flat if B is flat as an A-module.

Recall that we have seen that free modules are flat and projective modules are flat (as the consequence of higher Tor vanish and the LES). Moreover, recall that localization is exact, therefore we know that $B \to S^{-1}B$ for any multiplicative system is exact. Some easy examples are: since any module over a field is free, then any module over a field is flat. Since for any ring A, the polynomial ring $A[x_1, \cdots, x_n]$ is also free, then $A \to A[x_1, \cdots, x_n]$ is free.

Example 1.1. Consider the ring map:

$$\mathbb{Q}[x] \to \mathbb{Q}[y], \quad x \mapsto y^2$$

Recall that this is the double cover of the affine line, let's show that this is flat as this will be an important example later.

I claim that $\mathbb{Q}[y]$ is free over $\mathbb{Q}[x]$. Consider the map:

$$\mathbb{Q}[x] \oplus \mathbb{Q}[x] \mapsto \mathbb{Q}[y]$$

where the first 1 maps to 1 and the second 1 maps to y. Then consider the image of $(f(x), g(x))$ we know that it's image is:

$$f(y^2) + yg(y^2)$$

Notice that the degree of each term in these two polynomials $f(y^2)$ and $yg(y^2)$ are either all even or all odd, therefore in order for the sum to be 0, the only way is that $f = g = 0$. This proves that this is an injective map. Notice that any $F(y)$, by separating the even and odd degrees in y, it can be written as the form of $f(y^2) + yg(y^2)$, therefore it's surjective, therefore Q$[y]$ is free.

Here is an important observation we will use later. Notice that if $a \in A$ is a non zero divisor and M is flat over A, then we know that the map $M \xrightarrow{\times a} M$ where you multiply every element in M with a is going to be injective. This is because this can be identified with the map we get by tensor $A \xrightarrow{\times a} A$ with M. Therefore if there is a non zero divisor such that $M \xrightarrow{\times a} M$ is not injective then M is not flat. Therefore this is usually used as a criterion for nonflatness.

Example 1.2. For another example, if A now is a PID, then we will show that a finitely generated module is free iff it's torsion-free iff it's flat.

Recall that we know that any finitely generated module M over A is isomorphic to:

$$A^{\oplus r} \oplus A/(f_1) \oplus A/(f_2) \oplus \cdots \oplus A/(f_r)$$

where $f_i | f_{i+1}$ and none of them is 0. Then we know that M is free iff it's torsion-free. We also know that if M is free it's flat, therefore now we assume M is flat.

Since A is a domain, then we know that any nonzero element is a nonzerodivisor, therefore we know that if M is not free, then for example we know that the following map is not injective:

$$M \xrightarrow{\times f_1} M$$

Therefore M is not flat, therefore flatness implies freeness.

Proposition 1.3. If $B \to A$ is a ring map and M is flat over B, then $M \otimes_B A$ is flat over A. **This is saying that flatness is preserved by base change.**

Proof. Notice that we have natural isomorphisms:

$$N \otimes_A (M \otimes_B A) \simeq N \otimes_B M$$

and $- \otimes_B M$ is exact. □

Proposition 1.4. If $B \to A$ is flat and M is flat over A then M is flat over B. This can be summarized to **flatness is preserved by composition**.

Proof. Notice that:

$$N \otimes_B M \simeq N \otimes_B (A \otimes_A M) \simeq (N \otimes_B A) \otimes_A M$$

Then we use the composition of exact functor is exact. □

One good property about flatness is that it can be checked stalk locally:

Proposition 1.5. An A-module M is flat iff M_p if flat over A_p for all prime ideals p. (We can easily prove it suffices to check only all maximal ideals, but we don't need it.)

Proof. If M is flat over A, then M_p is flat over A_p is just the result of M_p is isomorphic $M \otimes_A A_p$ and flatness is preserved by base change.

Now if M_p is flat over A_p for all p, to check flatness of M, we only need to recall that injectiveness can be check stalk locally. □

Now we define flatness on the level of schemes:

Definition 1.6. Let X be a scheme and $\mathcal{F}$ a quasicoherent sheaf on X. We say that $\mathcal{F}$ is **flat at** $p \in X$ if $\mathcal{F}_p$ is flat over $\mathcal{O}_{X,p}$. We say that $\mathcal{F}$ is **flat over** X iff it's flat at any point $p \in X$.

Since $\mathcal{F}$ is quasicoherent, we can check this on an affine cover of X. That is to say that is $\widetilde{M}$ is flat over $SpecA$ for an A-module M iff M is flat over A.

Definition 1.7. Let $\pi : X \to Y$ be a morphism of schemes, we say that it's **flat at** p if $\mathcal{O}_{X,p}$ is flat over $\mathcal{O}_{Y,\pi(p)}$. We will say that π is **flat over** Y if it's flat at all $p \in X$.

These two definitions can actually be made into one definition:

Definition 1.8. Let $\pi : X \to Y$ be a morphism of scheme and $\mathcal{F}$ a quasicoherent sheaf over X. We will say that $\mathcal{F}$ is **flat over** Y **at** $p \in X$ if $\mathcal{F}_p$ is flat over $\mathcal{O}_{Y,f(p)}$. We say that $\mathcal{F}$ is **flat over** Y if it's flat over Y at any $p \in X$.

Then the two definitions above are just when π is the identity morphism of X and when $\mathcal{F} = \mathcal{O}_X$. And you should notice that this can be both checked locally on Y and locally on X.

Example 1.9. One easy example of a class of flat morphisms is the **open embedding**. As we know that the corresponding morphism of the stalks is just the identity.

Example 1.10. Let's see what this notion of flatness looks like in the affine case.

Let $B \to A$ be a ring map, then I claim this ring map is flat iff $SpecA \to SpecB$ is flat. More generally, if M is an A-module, then M if flat over B iff $\widetilde{M}$ is flat over $SpecB$.

This is most easily proved by consider it's contrapositive. First suppose that M is not flat over B, then there will be an exact sequence of B-modules:

$$0 \to N' \to N \to N'' \to 0$$

such that:

$$N' \otimes_B M \to N \otimes_B M$$

is not injective. Notice that this map can also be viewed as a map of A-modules, then we know that it's not injective, therefore there will be a prime p in A such that:

$$N' \otimes_B M_p \to N \otimes_B M_p$$

is not injective. But note that the above map is identified with:

$$N'_q \otimes_{B_q} M_p \to N_q \otimes_{B_q} M_p$$

and this is not injective where q is the inverse image of p. This is because we have a natural isomorphism:

$$N \otimes_B A_p \simeq N \otimes_B B_q \otimes_{B_q} A_p \simeq N_q \otimes_{B_q} A_p$$

Notice that $N'_q \to N_q$ is injective, therefore M_p is not flat over B_q. Therefore $\widetilde{M}$ is not flat over $SpecB$.

Conversely, let's assume $\widetilde{M}$ is not flat over $SpecB$, we know that there will exist a prime in A such that M_p is not flat over B_q where q is the inverse image of p. Then we know that there is a SES of B_q-modules:

$$0 \to N' \to N \to N'' \to 0$$

such that $N' \otimes_{B_q} M_p \to N \otimes_{B_q} M_p$ is not injective. Using similar argument as above we know that:

$$N' \otimes_B M_p \to N \otimes_B M_p$$

is not injective, thus $N' \otimes_B M \to N \otimes_B M$ is not injective, we are done.

Corollary 1.11. If $\pi : X \to Y$ is an affine morphism and $\mathcal{F}$ is a quasicoherent sheaf over X, then $\mathcal{F}$ is flat over Y iff $\pi_*\mathcal{F}$ is flat over Y.

Proof. Since $\pi : X \to Y$ is affine and flatness can be checked affine locally, we know that we can assume that π is just $SpecA \to SpecB$ and $\mathcal{F} = \widetilde{M}$. Then we know that $\widetilde{M}$ is flat over $SpecB$ iff M if flat over B. However we know that the pushforward of $\widetilde{M}$ to $SpecB$ is just $\widetilde{M}$ where M is regarded as a B-module. □

Example 1.12. Let X be a scheme, then the canonical map $Spec\mathcal{O}_{X,p} \to X$ is flat. To check this, notice that we can assume X is just an affine scheme, say $SpecA$, then the morphism is just $SpecA_p \to SpecA$, then we know that $A \to A_p$ is flat.

The canonical projection $\mathbb{A}^n_A \to SpecA$ is flat, as we know that $A \to A[x_1, \cdots, x_n]$ is flat.

Similarly, if $\mathcal{F}$ is a locally free sheaf of finite rank over X, then $\mathbb{P}\mathcal{F} \to X$ is flat. We check it again locally on X, and we can assume this is just $\mathbb{P}^n_A \to SpecA$, this is because if we want to check it's flat over $p \in \mathbb{P}^n_A$, we only need to assume we work with the canonical projection $\mathbb{A}^n_A \to SpecA$, which is flat, therefore we are done.

Finally $Speck \to Speck[t]/(t^2)$ is not flat. Consider the inclusion:

$$(\bar{t}) \hookrightarrow k[t]/(t^2)$$

Let's tensor with k, which is:

$$(\bar{t}) \otimes_{k[t]/(t^2)} k \to k[t]/(t^2) \otimes_{k[t]/(t^2)} k$$

Notice that the first term is just $(t)/(\bar{t})^2$, but notice that $\bar{t}^2 = 0$, therefore we know that we get:

$$(\bar{t}) \to k, \quad \bar{t} \mapsto t \cdot 1 = 0$$

which is not injective. Therefore we are done.

Proposition 1.13. If $\pi : X \to Y$ is a flat morphism, then $\pi^* : \mathrm{Qcoh}_Y \to \mathrm{Qcoh}_X$ is exact.

Proof. Let:

$$0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{F}'' \to 0$$

be an exact sequence of quasicoherent sheaves over X. We want to show that $\pi^*\mathcal{F}' \to \pi^*\mathcal{F}$ is exact. Recall that if $p \in X$, we know that the induced morphism on stalk $(\pi^*\mathcal{F}')_p \to (\pi^*\mathcal{F})_p$ can be identified with:

$$\mathcal{F}'_{\pi(p)} \otimes_{\mathcal{O}_{Y,\pi(p)}} \mathcal{O}_{X,p} \to \mathcal{F}_{\pi(p)} \otimes_{\mathcal{O}_{Y,\pi(p)}} \mathcal{O}_{X,p}$$

where $\mathcal{F}'_{\pi(p)} \to \mathcal{F}_{\pi(p)}$ is injective. Then since $\mathcal{O}_{X,p}$ is flat over $\mathcal{O}_{Y,\pi(p)}$, we know that after we tensor with $\mathcal{O}_{X,p}$, it's still injective. □

Proposition 1.14. Suppose that $\pi : X \to Y$ is a morphism of scheme and $\mathcal{F}$ is a quasicoherent sheaf on X flat over Y. Let $\psi : Y \to Z$ be a flat morphism, then $\mathcal{F}$ is flat over Z.

Proof. We need to show that for each $p \in X$, $\mathcal{F}_p$ is flat over $\mathcal{O}_{Z,\psi\pi(p)}$ where we have a ring map:

$$\mathcal{O}_{Z,\psi\pi(p)} \to \mathcal{O}_{Y,\pi(p)} \to \mathcal{O}_{X,p}$$

Since we know that ψ is flat, we know that $\mathcal{O}_{Z,\psi\pi(p)} \to \mathcal{O}_{Y,\pi(p)}$ is flat. Moreover $\mathcal{F}_p$ is flat over $\mathcal{O}_{Y,\pi(p)}$, then we know from the fact that flat module is preserved by composition, we are done. □

Proposition 1.15. Suppose that $\pi : X \to Y$ is a morphism and $\mathcal{F}$ is a quasicoherent sheaf on X that is flat over Y. Let $\rho : Y' \to Y$ be any morphism of schemes with $\rho' : X \times_Y Y' \to X$ being the induced morphism. Then $(\rho')^*\mathcal{F}$ is flat over Y'.

Proof. The diagram looks like:

$$\begin{tikzcd}
(\rho')^*\mathcal{F} \arrow[d, dash] & \mathcal{F} \arrow[d, dash] \\
X \times_Y Y' \arrow[r, "\rho'"] \arrow[d, "\pi'"'] & X \arrow[d, "\pi"] \\
Y' \arrow[r, "\rho"'] & Y
\end{tikzcd}$$

We need to show that for each $p \in X \times_Y Y'$, $(\rho')^*\mathcal{F}_p$ is flat over $\mathcal{O}_{Y',\pi'(p)}$. Recall that this Cartesian diagram induces a diagram:

$$\begin{tikzcd}
(\rho')^*\mathcal{F} \arrow[d, dash] & \mathcal{F} \arrow[d, dash] \\
\mathcal{O}_{X\times_Y Y',p} & \mathcal{O}_{X,\rho'(p)} \arrow[l, "\rho'"'] \\
\mathcal{O}_{Y',\pi'(p)} \arrow[u, "\pi'"] & \mathcal{O}_{Y,q} \arrow[l, "\rho"] \arrow[u, "\pi"']
\end{tikzcd}$$

where $q \in Y$ is $\pi\rho'(p) = \rho\pi'(p)$. Recall that $\mathcal{O}_{X\times_Y Y',p}$ is a further localization of $\mathcal{O}_{X,\rho'(p)} \otimes_{\mathcal{O}_{Y,q}} \mathcal{O}_{Y',\pi'(p)}$ (see the remark below), which is flat. Then We have the following diagram:

$$\begin{tikzcd}
\mathcal{O}_{X,\rho'(p)} \otimes_{\mathcal{O}_{Y,q}} \mathcal{O}_{Y',\pi'(p)} \arrow[rr, dashed] & & (\rho')^*\mathcal{F} \arrow[d, dash] & \mathcal{F} \arrow[d, dash] \\
& & \mathcal{O}_{X\times_Y Y',p} & \mathcal{O}_{X,\rho'(p)} \arrow[l, "\rho'"'] \arrow[ull] \\
& & \mathcal{O}_{Y',\pi'(p)} \arrow[u, "\pi'"] \arrow[uull] & \mathcal{O}_{Y,q} \arrow[l, "\rho"] \arrow[u, "\pi"']
\end{tikzcd}$$

Here the dotted map is induced by the universal property of tensor products of algebra, which is identified with a localization map. (Again, see the remark below) We know that the stalk of $(\rho')^*\mathcal{F}$ at p is $\mathcal{F}_{\rho'(p)} \otimes_{\mathcal{O}_{X,\rho'(p)}} \mathcal{O}_{X\times_Y Y',p}$. Here we know that: $\mathcal{F}_{\rho'(p)} \otimes_{\mathcal{O}_{Y,q}} \mathcal{O}_{Y',\pi'(p)}$ is flat over $\mathcal{O}_{Y',\pi'(p)}$, then $\mathcal{O}_{X\times_Y Y',p} \to \mathcal{O}_{X,\rho'(p)} \otimes_{\mathcal{O}_{Y,q}} \mathcal{O}_{Y',\pi'(p)}$ is flat. Moreover, we can think of $\mathcal{F}_{\rho'(p)} \otimes_{\mathcal{O}_{Y,q}} \mathcal{O}_{Y',\pi'(p)}$ as:

$$\mathcal{F}_{\rho'(p)} \otimes_{\mathcal{O}_{X,\rho'(p)}} \mathcal{O}_{X,\rho'(p)} \otimes_{\mathcal{O}_{Y,q}} \mathcal{O}_{Y',\pi'(p)}$$

Then we know that since $\mathcal{O}_{X,\rho'(p)} \otimes_{\mathcal{O}_{Y,q}} \mathcal{O}_{Y',\pi'(p)} \to \mathcal{O}_{X\times_Y Y',p}$ is flat, then consider:

$$\begin{tikzcd}
& \mathcal{F}_{\rho'(p)} \otimes_{\mathcal{O}_{Y,q}} \mathcal{O}_{Y',\pi'(p)} \arrow[d, dash] & \mathcal{F}_{\rho'(p)} \otimes_{\mathcal{O}_{Y,q}} \mathcal{O}_{X\times_Y Y',p} \arrow[d, dash] \\
\mathcal{O}_{Y',\pi'(p)} \arrow[r] & \mathcal{O}_{X,\rho'(p)} \otimes_{\mathcal{O}_{Y,q}} \mathcal{O}_{Y',\pi'(p)} \arrow[r] & \mathcal{O}_{X\times_Y Y',p}
\end{tikzcd}$$

We know that $\mathcal{F}_{\rho'(p)} \otimes_{\mathcal{O}_{Y,q}} \mathcal{O}_{X\times_Y Y',p}$ is flat over $\mathcal{O}_{X\times_Y Y',p}$ as $\mathcal{O}_{X,\rho'(p)} \otimes_{\mathcal{O}_{Y,q}} \mathcal{O}_{Y',\pi'(p)} \to \mathcal{O}_{X\times_Y Y',p}$ is flat. Then once we prove $\mathcal{O}_{Y',\pi'(p)} \to \mathcal{O}_{X\times_Y Y',p}$ is flat, we are done, i.e. we need to prove that $\mathcal{O}_{Y',\pi'(p)} \to \mathcal{O}_{X,\rho'(p)} \otimes_{\mathcal{O}_{Y,q}} \mathcal{O}_{Y',\pi'(p)}$ is flat, but this is again the fact that flatness is preserved by change of the base ring.

In conclusion, this proposition is proved by noting $\mathcal{O}_{X,\rho'(p)} \otimes_{\mathcal{O}_{Y,q}} \mathcal{O}_{Y',\pi'(p)} \to \mathcal{O}_{X\times_Y Y',p}$ is flat and use flatness is preserved by change of the base ring. $\square$

Remark 1.16. We see a similar remark in the note series of etale cohomology about why the stalk of the pullback localizes to the tensor product of algebras. Here we prove it again just for completeness.

We only need to consider the affine case, and we can assume that we are considering a diagram:

$$\begin{tikzcd} Spec A \otimes_C B \arrow[r, "\rho'"'] \arrow[d, "\pi'"] & Spec A \arrow[d, "\pi"'] \\ Spec B \arrow[r, "\rho"] & Spec C \end{tikzcd}$$

suppose that p is a prime ideal in $A \otimes_C B$, again we use $\rho'(p), \pi'(p)$ and q. Notice that there is a multiplicative system S in $A \otimes_C B$, which consists of element $x \otimes y$ where $x \notin \pi'(p)$ and $y \notin \rho'(q)$ (you can easily check this is indeed a multiplicative system). Then notice that this is contained in the multiplicative system $A \otimes_C B - p$.

Then notice that we have a natural isomorphism of $A \otimes_C B$-algebras: $S^{-1}(A \otimes_C B) \simeq A_{\rho'(p)} \otimes_{C_q} B_{\pi'(p)}$ where the isomorphisms are indicated below:

$$\begin{tikzcd} & A \otimes_C B \arrow[dl] \arrow[dr] & \\ S^{-1}(A \otimes_C B) \arrow[rr, leftrightarrow, dashed] & & A_{\rho'(p)} \otimes_{C_q} B_{\pi'(p)} \end{tikzcd}$$

The map from $S^{-1}(A \otimes_C B)$ is given by the universal property of localization. It's inverse is given by the universal property of tensors of C_q-algebras. We first define:

$$A_{\rho'(p)} \to S^{-1}(A \otimes_C B), \quad \frac{x}{y} \mapsto \frac{x \otimes 1}{y \otimes 1}$$

Similarly we define:

$$B_{\pi'(p)} \to S^{-1}(A \otimes_C B), \quad \frac{x}{y} \mapsto \frac{1 \otimes x}{1 \otimes y}$$

Then you can check that after precomposing $C_q \to A_{\rho'(p)}$ and $C_q \to B_{\pi'(p)}$, they agree, therefore we are done.

In particular, taking $\mathcal{F}$ to be $\mathcal{O}_X$, you get the fact that flat morphsims are preserved by base change.

Remark 1.17. We have already said that flat morphisms can be checked locally both on the source and target. The previous two propositions shows that flat morphisms also behave well with respect to composition and base change. Therefore flat morphism is a nice class.

Another good property of flat morphism is that it will send associated points to associated points:

Proposition 1.18. Suppose that $\pi : X \to Y$ is a flat morphism of locally Noetherian schemes, then any associated point of X will be mapped to an associated point of Y. More generally, if $\mathcal{F}$ is a coherent sheaf of X (still with the Noetherian hypopthesis) flat over Y, then any morphism $\pi : X \to Y$ maps associated point of $\mathcal{F}$ to associated point of Y.

Proof. First, suppose that $\pi^\sharp : (B, n) \to (A, m)$ is a local morphism of Noetherian local rings, such that A is flat over B. If n in B is not an associated prime of B, then I claim there is an $f \in n$ such that it's not in any associated prime of B and hence a nonzerodivisor. To see why, we use prime avoidance lemma, since n is not an associated prime, then we know that if all elements in n are zerodivisors, it's contained in the finitely many associated primes, then prime avoidance tells you that n is such an associated prime, a contradiction. Then since $B \to A$ is flat, we know that consider tensor $B \xrightarrow{\times f} B$ with A, which gives $A \xrightarrow{\times \pi^\sharp f} A$, which is injective, therefore $\pi^\sharp f \in m$ is also a nonzerodivisor. Therefore m is not an associated prime of A.

Now consider $\pi : X \to Y$, if $x \in X$ is an associated point, we want to show that $y = \pi(x)$ is also an associated point. Consider the induced map $\mathcal{O}_{Y,y} \to \mathcal{O}_{X,x}$, recall that we know that the maximal ideal of $\mathcal{O}_{X,x}$ is an associated prime, as it's preserved under localization. Recall that the notion of associated primes commutes with localization, then y is an associated prime of Y iff the maximal ideal of $\mathcal{O}_{Y,y}$ is an associated prime. But we proved that if y is not, then x is not. Since x is, then y is.

Immediately, this argument can be extended to coherent sheaf $\mathcal{F}$ on Y. Here we will use the fact that if $\pi^\sharp : (B, n) \to (A, m)$ is a local map between local Noetherian rings and suppose that M is a finitely generated

module over B that is flat with n is not an associated point of B. Then we know that there is an $f \in n$ such that $B \xrightarrow{\times f} B$ is injective, then $M \xrightarrow{\times f} M$ is also injective since M is flat, therefore we know that m is not an associated point of $\mathcal{F}$. The rest of the proof looks similar. $\square$

The above result can help us determine what morphisms are not flat:

Example 1.19. Let's first show that the morphism $Speck[x,y]/(xy) \to Speck[x]$, i.e. the morphism that project the x-axis and y-axis down to the x-axis is not flat. Notice that (x) and (y) are associated primes of $Speck[x,y]/(xy)$, while (x) is mapped to (x) in $Speck[x]$, which is not an associated prime since x is not an zerodivisor.

Then consider the morphism $Speck[x,y]/(y^2,xy) \to Speck[x]$, which you can think of as project the x-axis with a fuzz at the origin to the reduced x-axis. Notice that again we know (x) is an associated prime of $Speck[x,y]/(y^2,xy)$, whose image is (x), not an associated prime.

Finally, consider the canonical map $\mathrm{Bl}_{(0,0)}\mathbb{A}^2_k \to \mathbb{A}^2_k$. I claim that this is not flat. To see why, consider the following pullback diagram:

$$\begin{tikzcd}
Projk[X,Y,x]/(xY) \arrow[r] \arrow[d] & Projk[X,Y,x,y]/(Xy-xY) \arrow[d] \\
Speck[x] \arrow[r] & Speck[x,y]
\end{tikzcd}$$

Since flat morphism is preserved by base change, we know that if $\mathrm{Bl}_{(0,0)}\mathbb{A}^2_k \to \mathbb{A}^2_k$ is flat, then $Projk[X,Y,x]/(xY) \to Speck[x]$ is flat. Since flatness is local on the source, we know that the following morphism should be flat:

$$Speck[\frac{Y}{X},x]/(x\frac{Y}{X}) \to Speck[x]$$

But we already know this is not flat from the first example.

Remark 1.20. Case 1 and 3 in the previous example will be quite important as they are examples showing that there is a jumping in the fiber dimension. In the case of $Speck[x,y]/(xy) \to Speck[x]$, we know that the fiber of the origin is dimension 1 while all the other fibers of k-points is of dimension 0. Similarly, in the morphism $\mathrm{Bl}_{(0,0)}\mathbb{A}^2_k \to \mathbb{A}^2_k$, the fiber of $(0,0)$ is $\mathbb{P}^1$ which is dimension 1 while other fibers of k-points have dimension 0.

We will later show that flat morphisms won't have such jumpings in fiber dimensions.

Then let's show that cohomology plays well with flat base change:

Theorem 1.21. Suppose we have the following Cartesian diagram:

$$\begin{tikzcd}
X' \arrow[r, "\rho'"] \arrow[d, "\pi'"'] & X \arrow[d, "\pi"] \\
Y' \arrow[r, "\rho"'] & Y
\end{tikzcd}$$

let π be quasicompact and separated. Suppose that ρ is flat and $\mathcal{F}$ is a quasicoherent sheaf on X, then the natural push-pull morphism:

$$\rho^*(\mathrm{R}^i\pi_*\mathcal{F}) \to \mathrm{R}^i\pi'_*(\rho'^*\mathcal{F})$$

is an isomorphism.

Proof. Recall that we assumed that π and quasicompact and separated, then so is π'. Then Cech cohomology and Cech higher pushforward makes sense. Also recall that we have proved a similar result when ρ is affine. We defined the push-pull morphism above using the Cech definition and FHHF theorem. So to check this is an isomorphism, we only need to do it locally.

Let's review again how we defined push-pull map locally. Notice that if we pick an affine open of Y, say $SpecC$. Then we can cover the inverse image $\rho^{-1}SpecC$ using affines. Let one of be $SpecB$. Now we want to give a map:

$$\rho^*(\mathrm{R}^i\pi_*\mathcal{F})(SpecB) \to \mathrm{R}^i\pi'_*(\rho'^*\mathcal{F})(SpecB)$$

which will be a map:

$$H^i(X, \mathcal{F}) \otimes_C B \to H^i(X', \rho'^* \mathcal{F})$$

Therefore we can assume that ρ is affine. Recall that we need to choose an affine open cover of X to compute $H^i(X, \mathcal{F})$, and since ρ' is affine, we know that the inverse images in X' can be used to compute $H^i(X', \rho'^* \mathcal{F})$. Then I will let you see that the complex we use to compute the cohomology of $\rho'^* \mathcal{F}$ is the complex we use to compute the cohomology of $\mathcal{F}$ tensored with $\otimes_C B$, then the HFFH theorem gives you the desired morphism.

Now since B is flat over C, then tensor $\otimes_C B$ is exact, then the HFFH map is going to be an isomorphism, we are done. $\square$

Remark 1.22. If we only want to prove $i = 0$, then we can just assume π is quasiseparated instead of separated since separated is only intended for the Cech cohomology to make sense. If we only care about the usual pushforward, we only need qcqs to make sure the pushforward is still quasicoherent and now we only care about the 0-th cohomology of the complex, so we don't care if the intersection is affine or not.

Remark 1.23. Recall that if $Y' \to Y$ is $Spec\mathcal{O}_{Y,q} \to Y$ where q is the generic point of a reduced component of Y (that is to make sure that $\mathcal{O}_{Y,q}$ is a field), then we know that since $Y' \to Y$ is flat, we can apply the above result and get an isomorphism between the stalk of the i-th higher pushforward and the i-th cohomology of the sheaf $\mathcal{F}$ on the fiber of the generic point.

This is the first example something that is important: understanding cohomology of quasicoherent sheaves on fibers through the stalk of the higher pushforward. This point of view provide the higher pushforward very important geometric meaning. In the next chapter, we will discuss when and how an when the cohomology a flat quasicoherent sheaf commutes with any base change.

Now we will study how closed subschemes and the corresponding ideal sheaf pullback under flat morphisms. Recall that if $i : X \hookrightarrow Y$ is a closed embedding and $\mu : Y' \to Y$ is any morphism, then we can form the fiber product $i' : X' \to Y'$ which is also a closed embedding:

$$\begin{tikzcd} X' \arrow[r] \arrow[d, hook, "i'"] & X \arrow[d, hook, "i"] \\ Y' \arrow[r, "\mu"] & Y \end{tikzcd}$$

We know that as closed embeddings, i, i' correspond to quasicoherent ideal sheaves $\mathcal{I}_{X/Y}$ and $\mathcal{I}_{X'/Y'}$. Consider the exact sequence of the closed embedding i:

$$0 \to \mathcal{I}_{X/Y} \to \mathcal{O}_Y \to i_* \mathcal{O}_X \to 0$$

We can apply the pullback functor μ^* to get a right exact sequence:

$$\mu^* \mathcal{I}_{X/Y} \to \mu^* \mathcal{O}_Y \to \mu^* i_* \mathcal{O}_X \to 0$$

Notice that we also have the exact sequence corresponding to the closed embedding i', and we will have:

$$\begin{tikzcd} & \mu^* \mathcal{I}_{X/Y} \arrow[r] & \mu^* \mathcal{O}_Y \arrow[r] \arrow[d, "\simeq"] & \mu^* i_* \mathcal{O}_X \arrow[r] & 0 \\ 0 \arrow[r] & \mathcal{I}_{X'/Y'} \arrow[r] & \mathcal{O}_{Y'} \arrow[r] & i'_* \mathcal{O}_{X'} \arrow[r] & 0 \end{tikzcd}$$

where the isomorphism between $\mu^* \mathcal{O}_Y \to \mathcal{O}_{Y'}$ is the canonical one. Then notice that if $Y = SpecA$ then we know that the exact sequence of i is:

$$0 \to I \to A \to A/I \to 0$$

Say we pick $SpecB$ in $\mu^{-1} SpecA$, then we know that the pullback morphism on $SpecB$ is:

$$I \otimes_A B \to B \to A/I \otimes_A B$$

Then you can check that $\mu^* \mathcal{I}_{X/Y} \to \mu^* \mathcal{O}_Y \to \mathcal{O}_{Y'} \to i'_* \mathcal{O}_{X'}$ on $SpecB$ is:

$$I \otimes_A B \to B \to B \to B/IB$$

Therefore $\mu^*\mathcal{I}_{X/Y} \to \mu^*\mathcal{O}_Y \to \mathcal{O}_{Y'} \to i'_*\mathcal{O}_{X'}$ is 0, then the universal property cokernel gives a canonical map $\mu^* i_*\mathcal{O}_X \to i'_*\mathcal{O}_{X'}$, and we know that affine locally it's just $A/I \otimes_A B \to B/IB$ and you can see that this an isomorphism, therefore we actually have the following diagram:

$$\begin{tikzcd}
 & \mu^*\mathcal{I}_{X/Y} \arrow[r] \arrow[d, dashed] & \mu^*\mathcal{O}_Y \arrow[r] \arrow[d, "\simeq"] & i_*\mathcal{O}_X \arrow[r] \arrow[d, "\simeq"] & 0 \\
0 \arrow[r] & \mathcal{I}_{X'/Y'} \arrow[r] & \mathcal{O}_{Y'} \arrow[r] & i'_*\mathcal{O}_{X'} \arrow[r] & 0
\end{tikzcd}$$

where the dotted morphism is induced by the universal property of kernel which affine locally looks like $I \otimes_A B \to IB$. This maps is clearly surjective and if the row on the top is exact, then we can use five lemma to show that the induced morphism is an isomorphism. Recall that when μ is flat, the pullback is exact, then in this case the induced morphism is indeed an isomorphism. We will later using the ideal theoretic criterion to prove that not only flat morphsim implies this above isomorphism, the above isomorphism also characterizes flatness.

Remark 1.24. As an example of what happens when μ is not flat, consider the map $Speck \to Speck[x]$ which maps $Speck$ to the origin. We know that this is a closed embedding and the exact sequence is:

$$0 \to (t) \to k[t] \to k \to 0$$

However, if we again pullback this along $Speck \to Speck[t]$, we get:

$$(t)/(t^2) = (t) \otimes_{k[t]} k[t]/(t) \to k[t]/(t) \to k \to 0$$

Notice that $(t)/(t^2) \to k[t]/(t)$ is $at \mapsto 0$, therefore not injective.

Moreover we know that the induced map between the pullback ideal sheaf and the ideal sheaf of the pullback is:

$$(t) \otimes_{k[t]} k[t]/(t) \to 0$$

this is clearly not an isomorphism.

Therefore we know that the ideal sheaf of the pullback closed embedding is the pullback of the ideal sheaf of the original closed embedding in a canonical way. We not only just have this:

Proposition 1.25. Let D be an effective Cartier divisor on Y and $\pi : X \to Y$ is a flat morphism, then the pullback of D by π is also an effective Cartier divisor of X.

Proof. Let $\mathcal{I}_D$ be the ideal sheaf of D. We just prove that the ideal sheaf corresponding to the pullback of D is just $\pi^*\mathcal{I}_D$. We want to check that locally it's generated by a nonzero divisor, I will let you see that we can assume that Y is $SpecA$ and $\mathcal{I}_D$ is just (f) where f in A is a nonzero divisor, say X is $SpecB$ we know that the ideal sheaf is (f) in B. Notice that to check f is a nonzerodivisor is to check $B \xrightarrow{\times f} B$ is injective, but this is the tensor of $A \xrightarrow{\times f} A$ with B, which is injective since B is flat over A. $\square$

Corollary 1.26. If $i : X \hookrightarrow Y$ is a regular embedding, then if $Y' \to Y$ is flat, then the pullback $i' : X' \hookrightarrow Y'$ is also a regular embedding.

Proof. Recall that to check that the pullback is still a regular embedding, it suffices to check affine locally, then again we can think that $X = SpecA/I, Y = SpecA, Y' = SpecB$ are affine and we can assume that I is generated by a regular sequence $f_1, \cdots, f_d$, then we know that we can factor this embedding as a sequence of effective Cartier divisors:

$$\begin{tikzcd}
\cdots \arrow[r] & SpecA/(f_1, f_2) \arrow[r, hook] & SpecA/(f_1) \arrow[r, hook] & SpecA \\
\cdots \arrow[r] & SpecB/(f_1, f_2) \arrow[r, hook] \arrow[u] & SpecB/(f_1) \arrow[r, hook] \arrow[u] & SpecB \arrow[u]
\end{tikzcd}$$

This implies that IB is also generated by a regular sequence since each embedding in the row at the bottom is also an effective Cartier divisor. $\square$

For later use, we here explore more on pullbacks of closed embeddings:

Proposition 1.27. Suppose that $\pi : X \to Y$ is a morphism and $Z \hookrightarrow Y$ is a closed embedding cut out by an ideal sheaf $\mathcal{I} \subset \mathcal{O}_Y$. Just for this exercise, we use $\pi^\flat\mathcal{I} \subset \mathcal{O}_X$ to denote the ideal sheaf of the pullback of $Z \hookrightarrow Y$ by π (so $\pi^*\mathcal{I} \to \pi^\flat\mathcal{I}$ is also surjective but not injective if π is not flat). Then $(\pi^\flat\mathcal{I})^n = \pi^\flat(\mathcal{I}^n)$.

Proof. Recall that the two quasicoherent sheaf are both ideal sheaves of $\mathcal{O}_X$. Then we only need to check that they are locally the same, therefore we can again assume $X = SpecA/I, Y = SpecA, Y' = SpecB$ are affine. Then we know that locally $\pi^\flat\mathcal{I}$ is just $\widetilde{IB} \hookrightarrow \widetilde{B}$ and $\pi^\flat(\mathcal{I}^n)$ is just $\widetilde{I^nB} \hookrightarrow \widetilde{B}$. By the definition of $(\pi^\flat\mathcal{I})^n$, locally on $SpecB$ it's $\widetilde{IB}^n = \widetilde{I^nB}$, therefore we are done. $\square$

Proposition 1.28. If we use the notation from above, and assume π is flat with $Y = \mathbb{A}^n_k$ and Z is the origin.

$$\begin{tikzcd} W \arrow[r] \arrow[d, hook] & Z = Speck \arrow[d, hook] \\ X \arrow[r, "\pi"'] & Y = \mathbb{A}^n_k \end{tikzcd}$$

Let $\mathcal{J} = \pi^*\mathcal{I}$ be the quasicoherent ideal sheaf of ideals on X cutting out W, the pullback of Z by X. Then the quasicoherent graded sheaf of $\mathcal{O}_X$ algebras $\oplus_{n\geqslant 0}\mathcal{J}^n/\mathcal{J}^{n+1}$ is isomorphic as $\mathcal{O}_X$-algebra to the pushforward of $\mathcal{O}_W[x_1, \cdots, x_n]$.

Proof. The multiplication in $\oplus_{n\geqslant 0}\mathcal{J}^n/\mathcal{J}^{n+1}$ can be defined by working affine locally and then check it is compatible with restricting to distinguished opens since $\oplus_{n\geqslant 0}\mathcal{J}^n/\mathcal{J}^{n+1}$ is quasicoherent. We assume that affine locally we have $\oplus_{n\geqslant 0}J^n/J^{n+1}$ and multiplication from J^r/J^{r+1} and J^m/J^{m+1} goes to J^{m+r}/J^{m+r+1}. You should check this is indeed well defined and compatible with localization to a distinguished open.

Then let's prove that $\mathcal{J}^r/\mathcal{J}^{r+1} \simeq \mathrm{Sym}^r(\mathcal{J}/\mathcal{J}^2)$. Note that Y is affine, therefore we know that each affine open $SpecB$ in X, we can assume it's in the inverse image of an affine open of Y, i.e. itself. We will use A to denote $k[x_1, \cdots, x_n]$. Let's still say that the image of x_i in B is x_i. Then we know that $\mathcal{J}$ locally looks like $J = (x_1, \cdots, x_n)$. Let $C = B/J$, we know that J^r/J^{r+1} are C-modules and I claim that we have a canonical C-module isomorphism:

$$J^r/J^{r+1} \simeq \oplus_{a_1+\cdots+a_n=r}C[x_1^{a_1}\cdots x_n^{a_n}]$$

where the right hand is the direct sum labeled by monomials of degree n in $x_1, \cdots, x_n$. Clearly we have a map from right to left defined by $x_{i_1}\cdots x_{i_n}$ to $\overline{x_{i_1}\cdots x_{i_n}}$. This is clearly surjective and we only need to show it's injective. To show this, we will show that if:

$$\sum_{a_1\cdots a_n=r} b_{a_1\cdots a_n}x_1^{a_1}\cdots x_n^{a_n} = 0$$

Then $b_{a_1\cdots a_r} \in J$. To see this, we will need the fact that $x_1, \cdots, x_n$ is a regular sequence. Moreover, this is not just any arbitrary regular sequence, since we know that in $k[x_1, \cdots, x_n]$, we can arbitrarily reorder $x_1, \cdots, x_n$ and it remains a regular sequence, therefore so is $(x_1, \cdots, x_n) \in B$. Let's first prove the case when $r = 2$, i.e. J/J^2 is $\oplus_{1\leqslant i\leqslant n}C[x_i]$. Then we first module by $(x_1, \cdots, x_{n+1})$, we get $b_n \in J$, then we proceed by induction. Then we know that $\mathrm{Sym}^r(J/J^2)$ is free over C and can be though of as the monomials of degree r with variables $x_1, \cdots, x_n$, and we have a canonical C-linear map:

$$\mathrm{Sym}^r(J/J^2) \to J^r/J^{r+1}$$

which is defined by the universal property of symmetric products and it's clearly surjective, therefore only injective is the issue. We know that under our identification of J/J^2 as $\oplus_{1\leqslant i\leqslant n}C[x_i]$, the above map is the obvious map:

$$\oplus_{a_1+\cdots+a_n=r}C[x_1^{a_1}\cdots x_n^{a_n}] \to J^r/J^{r+1}$$

We only need to show this is injective. Let's again prove this using induction on n. First, let's assume that there is only one variable, i.e. we only have x_1 and the result is trivial. Now suppose that this holds for $n-1$ and we want to prove for n, that is to say if:

$$\sum_{a_1+\cdots+a_n=r} b_{a_1\cdots a_n} x_1^{a_1}\cdots x_n^{a_n} \in J^{r+1}$$

we want $b_{a_1\cdots a_n} \in J$. Let's group these monimials in terms of exponents of x_n, that is:

$$\sum_{j=0}^{r} x_n^j \Big(\sum_{a_1+\cdots+a_{n-1}=r-j} b_{a_1\cdots a_{n-1}j} x_1^{a_2}\cdots x_{n-1}^{a_{n-1}} \Big) \in J^{r+1}$$

Let's modulo x_r, we know that $x_1, \cdots, x_{n-1}$ is still a regular sequence in $B/(x_r)$, and we know that:

$$\sum_{a_1+\cdots+a_{n-1}=n} b_{a_1a_2\cdots a_{n-1}0} x_1^{a_1}\cdots x_{n-1}^{a_{n-1}} \in J^{r+1}/(x_r)$$

By our inductive hypothesis, we know that $b_{0a_2\cdots a_n} \in J/(x_r)$, therefore we know that $b_{a_1a_2\cdots a_{n-1}0} \in J+(x_1) = J$. We can subtract these terms and the result are still in J^{n+1}, and now we have:

$$\sum_{j=1}^{r} x_n^j \Big(\sum_{a_1+\cdots+a_{n-1}=r-j} b_{a_1\cdots a_{n-1}j} x_2^{a_1}\cdots x_{n-1}^{a_{n-1}} \Big) \in J^{r+1}$$

Now we can factor out a x_n and we have $x_n \cdot z \in J^{n+1}$ where z is what is left after we factor out x_r. Notice that each term in z is in J^r therefore we know that $z \in J^r$, then we can use the same method to conclude that the coefficients of these terms with 0 exponent of x_n is in J. Then we can keep doing this and conclude all coefficients are in J. This concludes injectivity.

Finally you can check this is compatible with localizations to distinguished opens which finishes the isomorphism. Therefore:

$$\oplus_{n\geqslant 0}\mathcal{J}^n/\mathcal{J}^{n+1} \simeq \oplus_{n\geqslant 0}\mathrm{Sym}^n(\mathcal{J}/\mathcal{J}^2)$$

Therefore now we only need to show that:

$$\oplus_{n\geqslant 0}\mathrm{Sym}^n(\mathcal{J}/\mathcal{J}^2) \simeq \mathcal{O}_W[x_1, \cdots, x_n]$$

Notice that if we can prove $\mathcal{J}/\mathcal{J}^2$ is $\mathcal{O}_W[x_1] \oplus \cdots \mathcal{O}_W[x_n]$, then we are done. We already proved the local version above and notice that both are quasicoherent, and the isomorphism build affine locally obvious is compatible with restricting to a distinguished open, thus we are done. □

Proposition 1.29. Let $\pi : X \to Y$ be a flat morphism and $Z \hookrightarrow Y$ is a closed embedding. Then there is a canonical isomorphism:

$$(\mathrm{Bl}_Z Y) \times_Y X \simeq \mathrm{Bl}_{Z\times_Y X} X$$

Proof. Consider the following diagram:

$$\begin{tikzcd}
 & & E_Z Y \arrow[r, hook] \arrow[dd] & \mathrm{Bl}_Z Y \arrow[dd] \\
Z \times_Y (X \times_Y \mathrm{Bl}_Z Y) \arrow[urr, dashed] \arrow[r, hook] \arrow[dd] & X \times_Y \mathrm{Bl}_Z Y \arrow[ur] \arrow[dd] & & \\
 & & Z \arrow[r, hook] & Y \\
Z \times_Y X \arrow[urr] \arrow[r, hook] & X \arrow[urr, "\pi"'] & &
\end{tikzcd}$$

Here the dotted morphism is induced by the universal property, and you can see that all sides of the square are Cartesian. Then we know that since π is flat, so is $X \times_Y \mathrm{Bl}_Z Y \to \mathrm{Bl}_Z Y$. Then we know that since

the pullback of an effective Cartier divisor by a flat morphism is also an effective Cartier divisor, then $Z \times_Y (X \times_Y \mathrm{Bl}_Z Y) \to X \times_Y \mathrm{Bl}_Z Y$ is also an effective Cartier divisor. Then we need to check the universal property. Consider the following diagram:

$$\begin{tikzcd}
D \arrow[r, hook] \arrow[dr, dashed] \arrow[ddr] & W \arrow[drr, dashed] \arrow[ddr] & & \\
& & E_Z Y \arrow[r, hook] \arrow[d] & \mathrm{Bl}_Z Y \arrow[d] \\
& Z \times_Y (X \times_Y \mathrm{Bl}_Z Y) \arrow[ur] \arrow[r, hook] \arrow[d] & X \times_Y \mathrm{Bl}_Z Y \arrow[ur] \arrow[d] & \\
& & Z \arrow[r, hook] \arrow[d] & Y \\
& Z \times_Y X \arrow[ur] \arrow[r, hook] & X \arrow[ur, "\pi"'] &
\end{tikzcd}$$

where $D \hookrightarrow W$ is an effective Cartier divisor and the square $D, W, X, Z \times_Y X$ is Cartesian. The dotted morphisms are induced by the obvious universal properties of Cartesian squares which are all uniquely given. This suffice to check the universal property of blow-ups. $\square$

Example 1.30. Here is an example that tells you that without flatness, blow-up does not necessarily commute with base change.

Let's consider the blow-up of the plane at the origin and the following diagram:

$$\begin{tikzcd}
& E_{(0,0)}\mathbb{A}^2_k \arrow[r, hook] \arrow[d] & \mathrm{Bl}_{(0,0)}\mathbb{A}^2_k \arrow[d] \\
Projk[X,Y] \arrow[ur, dashed] \arrow[r] \arrow[d] & Projk[X,Y] \arrow[ur] \arrow[d] & \\
& Speck \arrow[r, hook] \arrow[d] & Speck[x,y] \\
Speck \arrow[ur] \arrow[r, hook] & Speck \arrow[ur] &
\end{tikzcd}$$

Clearly $Speck \to Speck[x, y]$ defined by $x, y \mapsto 0$ is not flat then the blow-up of $Speck$ at (0) should still be $Speck$, instead of $\mathbb{P}^1$.

2 Flatness through Tor

References